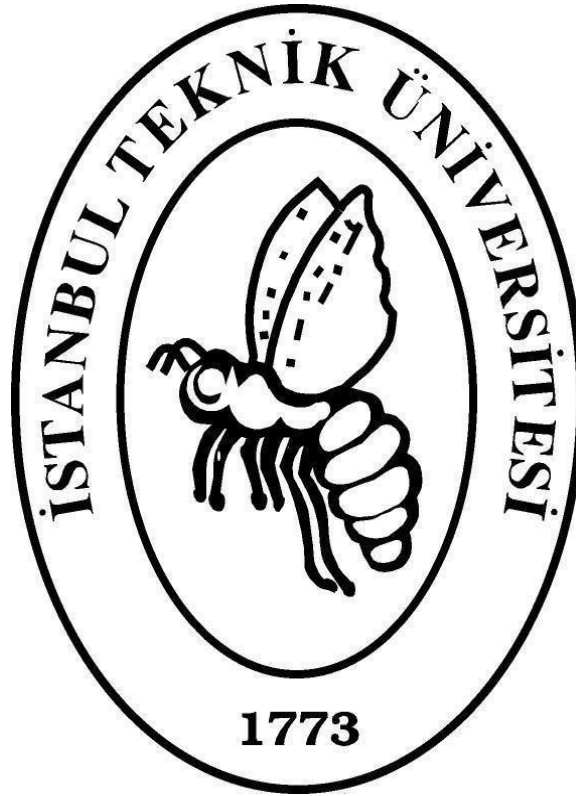




HOMEWORK I

ELK 346E – Data Acquisition and Embedded Systems

Student Number: 040200434

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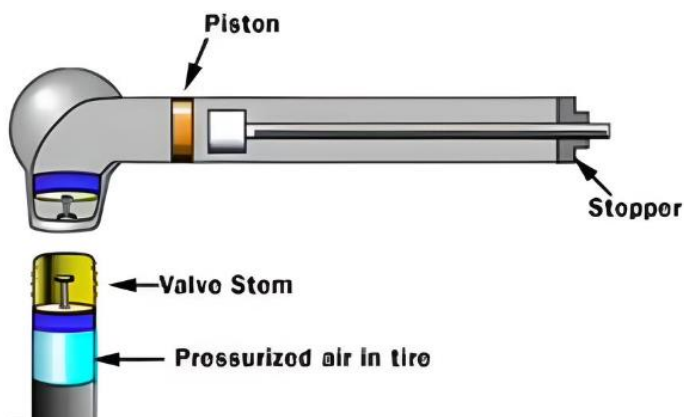
CRN: 22379

Course Instructor: Aydın Tarık ZENGİN

WEEK 1

- Describe the actual sensor in each of the following systems and the basic physical principle or law upon which each works.

1- Tire Pressure Gauge



The underlying principle of tire pressure gauges is the same as any pressure measuring device. They measure the force exerted by air on a unit area, simply pressure. When you connect a tire pressure gauge to your tire's valve stem, it forms a sealed system. The air pressure within the tire pushes against a sensing element in the gauge, causing it to move or change shape. This movement or deformation is then converted into a readable pressure value.

2- Toaster

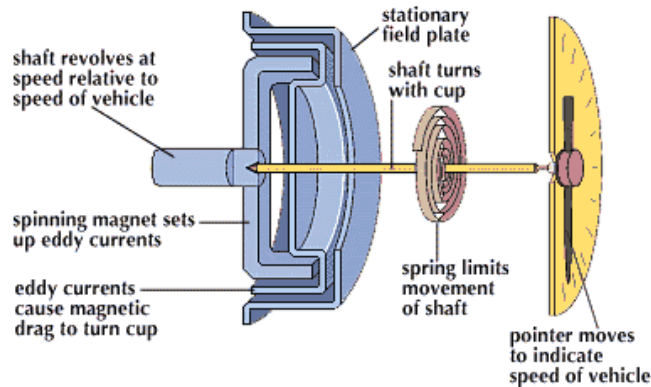


A thermostat contains a temperature-sensing component, often a bimetallic strip containing two different metals (aluminum, nickel, copper or tungsten). When exposed to heat, the difference in the linear coefficient of expansion of the two metals produces a mechanical bending motion. The strip carries electricity through the connected circuit, switching on the heating. As the strip heats up, one of the metals expands enough to open the circuit and switch off the heating, turning off the toaster.

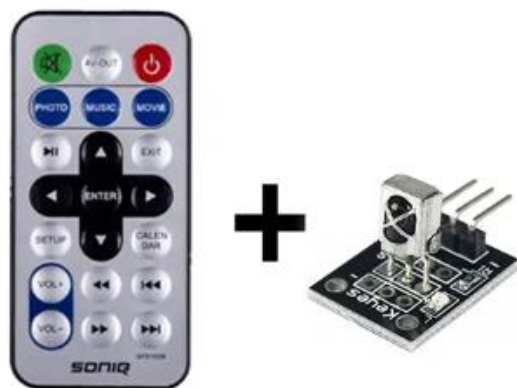
3- Car Mph/Kph Indicator

A mph/kph indicator uses a speedometer that indicates speed via a circular permanent magnet that rotates. This motion is initiated by a flexible shaft driven by gears at the rear of the transmission. The magnet turns within a movable metal cup made of a light nonmagnetic metal that is attached to the shaft carrying the indicating pointer. The magnetic circuit is completed

by a circular stationary field plate. As the magnet rotates, it exerts a magnetic drag on the movable cup that tends to turn it against the restraint of a spiral spring. The faster the magnet rotates, the greater the pull on the cup and the pointer.



4- Television Remote Control



The television remote control uses an infrared (IR) sensor. An IR remote (also called a transmitter) uses light to carry signals from the remote to the device it controls. An IR remote control (the transmitter) sends out pulses of infrared light that represent specific binary codes. These binary codes correspond to commands, such as Power On/Off and Volume Up. The IR receiver in the TV, stereo or other device decodes the pulses of light into the binary data (ones and zeroes) that the device's microprocessor can understand. The microprocessor then carries out the corresponding command.

5- Battery-Life Indicator On A Digital Camera

Battery life indicators on digital cameras typically use voltage sensors to estimate the remaining charge in the battery. There are two types of voltage sensors: those that measure voltage and those that measure current (or amperage). The first type is a voltage-sensitive resistor (VSR), while the second is an operational amplifier (op-amp).

VSRs are used in electronic circuits where you need to sense small changes in voltage. Thus, a battery life indicator on a digital camera uses this type of voltage sensor. This sensor mainly includes two circuits like a voltage divider & bridge circuit. The resistor in the circuit works

as a sensing element. The voltage can be separated into two resistors like a reference voltage & variable resistor to make a circuit of the voltage divider. A voltage supply is applied to this circuit. The output voltage can be decided by the resistance used in the circuit. So the voltage change can be amplified.

WEEK 2

- **Do research about the topics below for ADCs and DACs.**

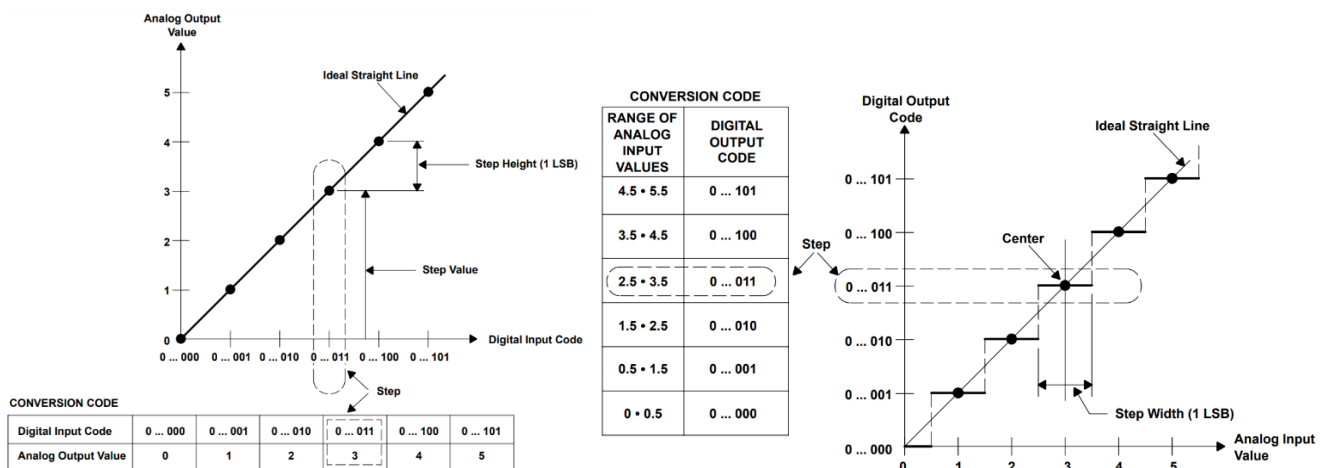
Since the analog scale is continuous, while the digital codes are discrete, there is a quantization process that introduces an error. Integral Nonlinearity(INL) and Differential Nonlinearity are two of these errors. As the number of discrete codes increases, the corresponding step width gets smaller and the transfer function approaches an ideal straight line. The steps are designed to have transitions such that the midpoint of each step corresponds to the point on this ideal line.

Resolution

The width of one step is defined as 1 LSB (one least significant bit) and this is often used as the reference unit for other quantities in the specification. Resolution defines the number of divisions or units of the full analog range. The resolution of an ADC is usually expressed as the number of bits in its digital output code. For example, an ADC with an n-bit resolution has 2^n possible digital codes which define 2^n step levels. However, since the first (zero) step and the last step are only one half of a full width, the full-scale range (FSR) is divided into $2^n - 1$ step widths. Hence, for an n-bit converter:

$$1 \text{ LSB} = \text{FSR} / (2^n - 1)$$

Conversions of an ideal DAC(on the left) and ADC(right)



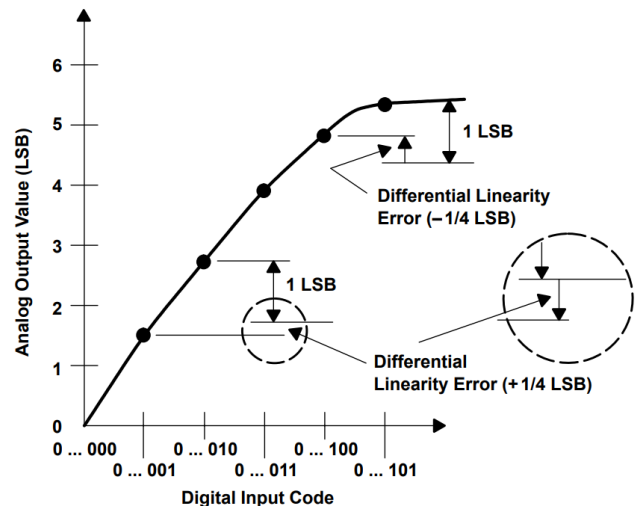
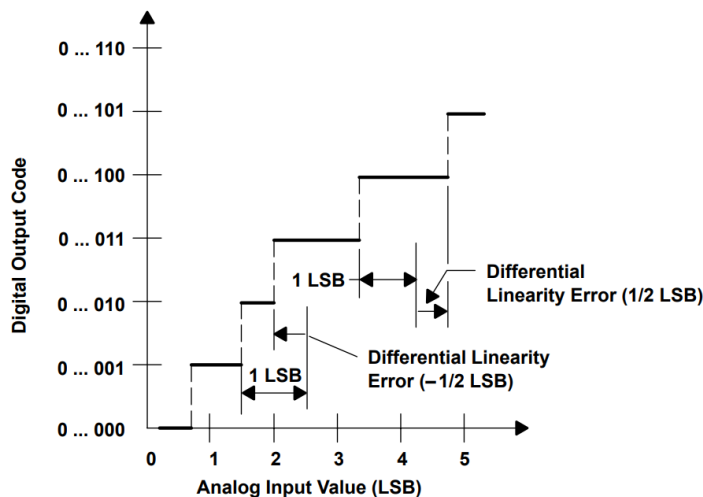
Conversion Time

The conversion time for an ADC is the time it takes for the ADC to sample the analog input signal, quantize it, and produce a digital output. For a DAC, conversion time refers to the time taken to convert a digital input into a corresponding analog output. This includes the time for the DAC to receive the digital input, process it, and produce the analog output voltage or current.

Differential Nonlinearity (DNL)

The differential nonlinearity error is the difference between an actual step width (for an ADC) or step height (for a DAC) and the ideal value of 1 LSB. Therefore if the step width or height is exactly 1 LSB, then the differential nonlinearity error is zero.

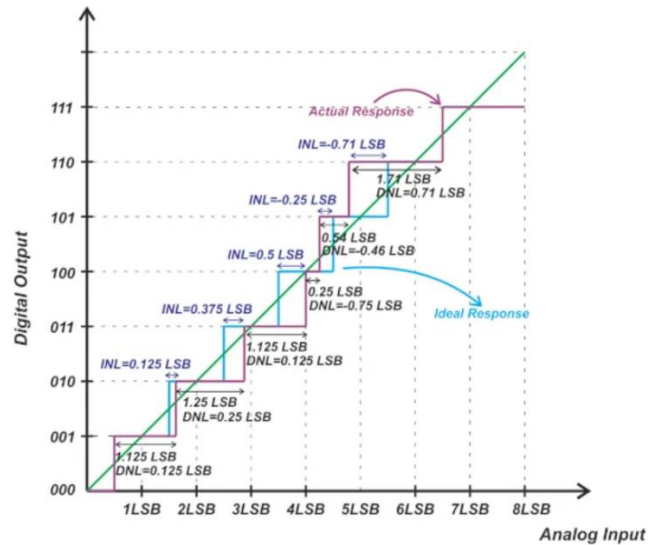
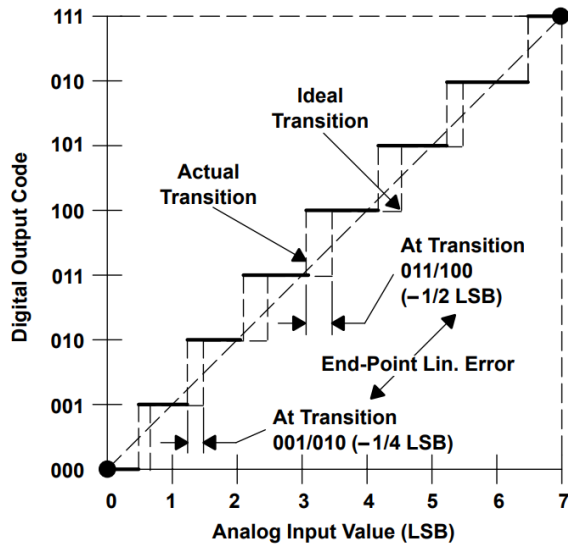
Differential Nonlinearity Error of a Linear ADC and DAC



Integral Nonlinearity (INL)

The integral nonlinearity error shown in Figure 6 (sometimes seen as simply linearity error) is the deviation of the values on the actual transfer function (output/input) from a straight line. For an ADC the deviations are measured at the transitions from one step to the next, and for the DAC they are measured at each step. The name integral nonlinearity derives from the fact that the summation of the differential nonlinearities from the bottom up to a particular step, determines the value of the integral nonlinearity at that step.

Calculation of Integral Nonlinearity Error of an ADC



WEEK 3

• What are the ADC/DAC capabilities of STM32 F1, F4 and F7 series? What are their resolutions? What's the sampling frequency range?

The information on STMicroelectronics' website on F1, F4, F7 series of MCUs is used in this part. The overview segment and the latest documentations on the relevant MCU was referenced. Also, resolution of each series is defined as the smallest incremental voltage that can be recognized and thus causes a change in the digital output. It is expressed as the number of bits output by the ADC. Therefore, an ADC which converts the analog signal to an N-bit digital value has a resolution of N bits.

STM32F1 Series

All devices offer two 12-bit ADCs. (Resolution is 12 bits.) Only high-density devices (256+KB Flash & 48+KB RAM) offer three 12-bit ADCs, two DACs. V_{DDA} and V_{SSA} (analog supply and ground voltages) are 2.0 to 3.6 V; min supply is 2.4 V when the ADC is used.

Two 12-bit ADC shares up to 16 external channels. Additional logic functions embedded in the ADC interface allow:

- Simultaneous sample and hold
- Interleaved sample and hold
- Single shunt

The ADC can be served by the DMA controller. An analog watchdog feature allows very precise monitoring of the converted voltage of one, some or all selected channels. An interrupt

is generated when the converted voltage is outside the programmed thresholds. The events generated by the general-purpose timers (TIMx) and the advanced-control timer (TIM1) can be internally connected to the ADC start trigger, injection trigger, and DMA trigger respectively, to allow the application to synchronize A/D conversion and timers. From the table below, we can understand the sampling frequency ranges from 0.6 to 14MHz.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Power supply	-	2.4	-	3.6	V
V_{REF+}	Positive reference voltage	-	2.4	-	V_{DDA}	V
I_{VREF}	Current on the V_{REF} input pin	-	-	160 ⁽¹⁾	220 ⁽¹⁾	μA
f_{ADC}	ADC clock frequency	-	0.6	-	14	MHz
$f_S^{(2)}$	Sampling rate	-	0.05	-	1	MHz
$f_{TRIG}^{(2)}$	External trigger frequency	$f_{ADC} = 14 \text{ MHz}$	-	-	823	kHz
			-	-	17	$1 / f_{ADC}$
$V_{AIN}^{(3)}$	Conversion voltage range		0 (V_{SSA} or V_{REF-} tied to ground)	-	V_{REF+}	V

STM32F4 Series

All devices offer one 12-bit, 2.4 MSPS (Megasamples per Second) ADC up to 16 channels in this series. V_{DDA} and V_{SSA} are 1.7 to 3.6 V. One 12-bit analog-to-digital converter is embedded and shares up to 16 external channels, performing conversions in the single-shot or scan mode. In scan mode, automatic conversion is performed on a selected group of analog inputs. The ADC can be served by the DMA controller. An analog watchdog feature allows very precise monitoring of the converted voltage of one, some or all selected channels. An interrupt is generated when the converted voltage is outside the programmed thresholds. To synchronize A/D conversion and timers, the ADCs could be triggered by any of TIM1, TIM2, TIM3, TIM4 or TIM5 timer.

From the documentation[2], we can obtain the following information about ADC in F4 series.

- When clock frequency is 30 MHz, and the resolution is 12-bit, our conversion time can range from 0.50 to 16.40 microseconds.
- The sampling frequency is up to 2 Msps using a 12-bit resolution single ADC

Operating power supply range	ADC operation	Maximum flash memory access frequency with no wait states (f_{Flashmax})	Maximum flash memory access frequency with wait states ⁽¹⁾⁽²⁾	I/O operation	Clock output frequency on I/O pins ⁽³⁾	Possible flash memory operations
$V_{\text{DD}} = 1.7$ to 2.1 V ⁽⁴⁾	Conversion time up to 1.2 Msps	16 MHz ⁽⁵⁾	100 MHz with 6 wait states	– No I/O compensation	up to 30 MHz	8-bit erase and program operations only
$V_{\text{DD}} = 2.1$ to 2.4 V	Conversion time up to 1.2 Msps	18 MHz	100 MHz with 5 wait states	– No I/O compensation	up to 30 MHz	16-bit erase and program operations
$V_{\text{DD}} = 2.4$ to 2.7 V	Conversion time up to 2.4 Msps	24 MHz	100 MHz with 4 wait states	– I/O compensation works	up to 50 MHz	16-bit erase and program operations
$V_{\text{DD}} = 2.7$ to 3.6 V ⁽⁶⁾	Conversion time up to 2.4 Msps	30 MHz	100 MHz with 3 wait states	– I/O compensation works	– up to 100 MHz when $V_{\text{DD}} = 3.0$ to 3.6 V – up to 50 MHz when $V_{\text{DD}} = 2.7$ to 3.0 V	32-bit erase and program operations

Table 72. ADC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Power supply	$V_{\text{DDA}} - V_{\text{REF+}} < 1.2$ V	1.7 ⁽¹⁾	-	3.6	V
$V_{\text{REF+}}$	Positive reference voltage		1.7 ⁽¹⁾	-	V_{DDA}	V
f_{ADC}	ADC clock frequency	$V_{\text{DDA}} = 1.7^{(1)}$ to 2.4 V	0.6	15	18	MHz
		$V_{\text{DDA}} = 2.4$ to 3.6 V	0.6	30	36	MHz
$f_{\text{TRIG}}^{(2)}$	External trigger frequency	$f_{\text{ADC}} = 30$ MHz, 12-bit resolution	-	-	1764	kHz
		-	-	-	17	$1/f_{\text{ADC}}$
V_{AIN}	Conversion voltage range ⁽³⁾	-	0 (V_{SSA} or $V_{\text{REF-}}$ tied to ground)	-	$V_{\text{REF+}}$	V
$R_{\text{AIN}}^{(2)}$	External input impedance	See Equation 1 for details	-	-	50	k Ω
$R_{\text{ADC}}^{(2)(4)}$	Sampling switch resistance	-	-	-	6	k Ω
$C_{\text{ADC}}^{(2)}$	Internal sample and hold capacitor	-	-	4	7	pF
$t_{\text{lat}}^{(2)}$	Injection trigger conversion latency	$f_{\text{ADC}} = 30$ MHz	-	-	0.100	μs
		-	-	-	3 ⁽⁵⁾	$1/f_{\text{ADC}}$
$t_{\text{latr}}^{(2)}$	Regular trigger conversion latency	$f_{\text{ADC}} = 30$ MHz	-	-	0.067	μs
		-	-	-	2 ⁽⁵⁾	$1/f_{\text{ADC}}$
$t_{\text{S}}^{(2)}$	Sampling time	$f_{\text{ADC}} = 30$ MHz	0.100	-	16	μs
		-	3	-	480	$1/f_{\text{ADC}}$

$t_{\text{CONV}}^{(2)}$	Total conversion time (including sampling time)	$f_{\text{ADC}} = 30 \text{ MHz}$ 12-bit resolution	0.50	-	16.40	μs
		$f_{\text{ADC}} = 30 \text{ MHz}$ 10-bit resolution	0.43	-	16.34	μs
		$f_{\text{ADC}} = 30 \text{ MHz}$ 8-bit resolution	0.37	-	16.27	μs
		$f_{\text{ADC}} = 30 \text{ MHz}$ 6-bit resolution	0.30	-	16.20	μs
		9 to 492 (t_s for sampling +n-bit resolution for successive approximation)				$1/f_{\text{ADC}}$

$f_s^{(2)}$	Sampling rate ($f_{\text{ADC}} = 30 \text{ MHz}$, and $t_s = 3 \text{ ADC cycles}$)	12-bit resolution Single ADC	-	-	2	MspS
		12-bit resolution Interleave Dual ADC mode	-	-	3.75	MspS
		12-bit resolution Interleave Triple ADC mode	-	-	6	MspS
$I_{\text{VREF+}}^{(2)}$	ADC V_{REF} DC current consumption in conversion mode	-	-	300	500	μA
$I_{\text{VDDA}}^{(2)}$	ADC V_{DDA} DC current consumption in conversion mode	-	-	1.6	1.8	mA

STM32F7 Series

The devices offer three 12-bit 2.4 MSPS ADC, up to 24 channels and two DACs. V_{DDA} and V_{SSA} are 1.7 to 3.6 V. Three 12-bit analog-to-digital converters are embedded and each ADC shares up to 16 external channels, performing conversions in the single-shot or scan mode. In scan mode, automatic conversion is performed on a selected group of analog inputs.

The ADC can be served by the DMA controller. An analog watchdog feature allows very precise monitoring of the converted voltage of one, some or all selected channels. An interrupt is generated when the converted voltage is outside the programmed thresholds. To synchronize A/D conversion and timers, the ADCs could be triggered by any of TIM1, TIM2, TIM3, TIM4, TIM5, or TIM8 timer.

From the documentation[3], we can obtain the following information about ADC and DAC in F7 series.

-When clock frequency is 30 MHz, and the resolution is 12-bit, our conversion time can range from 0.50 to 16.40 microseconds.

- The sampling frequency is up to 2.4 Msps using a 12-bit resolution single ADC

Table 72. ADC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Power supply	$V_{DDA} - V_{REF+} < 1.2 \text{ V}$	1.7 ⁽¹⁾	-	3.6	V
V_{REF+}	Positive reference voltage		1.7 ⁽¹⁾	-	V_{DDA}	V
f_{ADC}	ADC clock frequency	$V_{DDA} = 1.7^{(1)} \text{ to } 2.4 \text{ V}$	0.6	15	18	MHz
		$V_{DDA} = 2.4 \text{ to } 3.6 \text{ V}$	0.6	30	36	MHz
$f_{TRIG}^{(2)}$	External trigger frequency	$f_{ADC} = 30 \text{ MHz}$, 12-bit resolution	-	-	1764	kHz
		-	-	-	17	$1/f_{ADC}$
V_{AIN}	Conversion voltage range ⁽³⁾	-	0 (V_{SSA} or V_{REF-} tied to ground)	-	V_{REF+}	V
$R_{AIN}^{(2)}$	External input impedance	See Equation 1 for details	-	-	50	k Ω
$R_{ADC}^{(2)(4)}$	Sampling switch resistance	-	1.5	-	6	k Ω
$C_{ADC}^{(2)}$	Internal sample and hold capacitor	-	-	4	7	pF
$t_{lat}^{(2)}$	Injection trigger conversion latency	$f_{ADC} = 30 \text{ MHz}$	-	-	0.100	μs
		-	-	-	3 ⁽⁵⁾	$1/f_{ADC}$
$t_{latr}^{(2)}$	Regular trigger conversion latency	$f_{ADC} = 30 \text{ MHz}$	-	-	0.067	μs
		-	-	-	2 ⁽⁵⁾	$1/f_{ADC}$
$t_S^{(2)}$	Sampling time	$f_{ADC} = 30 \text{ MHz}$	0.100	-	16	μs
		-	3	-	480	$1/f_{ADC}$
$t_{CONV}^{(2)}$	Total conversion time (including sampling time)	$f_{ADC} = 30 \text{ MHz}$ 12-bit resolution	0.50	-	16.40	μs
		$f_{ADC} = 30 \text{ MHz}$ 10-bit resolution	0.43	-	16.34	μs
		$f_{ADC} = 30 \text{ MHz}$ 8-bit resolution	0.37	-	16.27	μs
		$f_{ADC} = 30 \text{ MHz}$ 6-bit resolution	0.30	-	16.20	μs
		9 to 492 (t_S for sampling +n-bit resolution for successive approximation)				$1/f_{ADC}$
$f_S^{(2)}$	Sampling rate ($f_{ADC} = 36 \text{ MHz}$, and $t_S = 3 \text{ ADC cycles}$)	12-bit resolution Single ADC	-	-	2.4	Mbps
		12-bit resolution Interleave Dual ADC mode	-	-	4.5	Mbps
		12-bit resolution Interleave Triple ADC mode	-	-	7.2	Mbps

● How to adjust sampling frequency?

The sampling frequency parameter is usable for direct mode and snapshot mode.

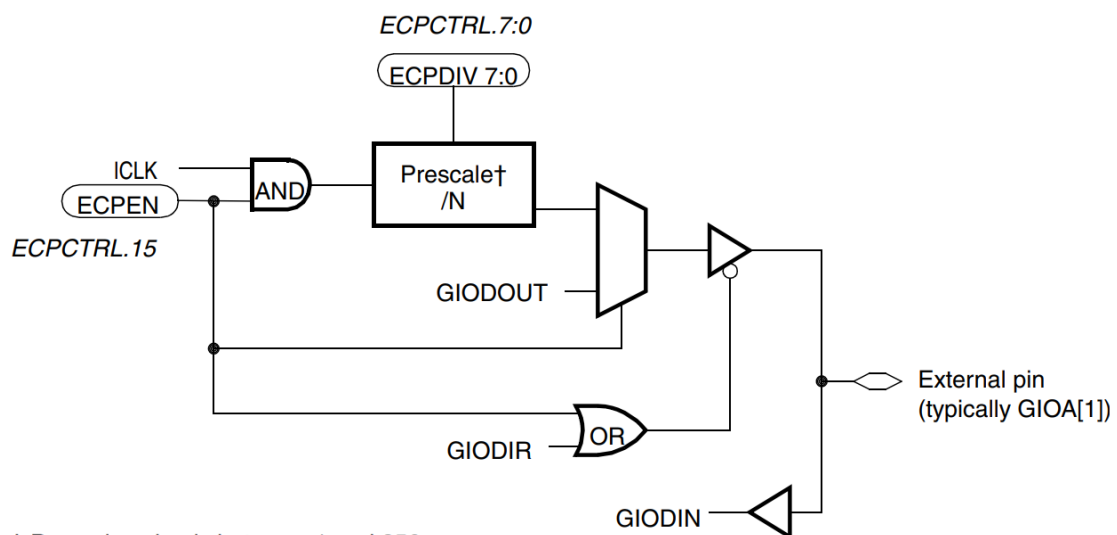
- In direct mode the sampling frequency is the frequency at which the variables are read out from the memory target.
- In the snapshot mode, the sampling frequency is the frequency at which the target buffer is emptied, the frequency at which the variables are read is defined in the embedded application by the user.

The sampling frequency is configured in the variable node. In “Acquisition parameters” part, the “Sampling frequency” field allows the sampling frequency to be set as needed.

● What is prescaler?

A prescaler divides down the clock signals used for the timer, giving reduced overflow rates. The rate can be set to a number of possible values. The prescaler takes the basic timer clock frequency (which may be the CPU clock frequency or may be some higher or lower frequency) and divides it by some value before feeding it to the timer, according to how the prescaler registers are configured. The prescaler values, referred to as prescales, that may be configured might be limited to a few fixed values (powers of 2), or they may be any integer value from 1 to 2^P , where P is the number of prescaler bits.

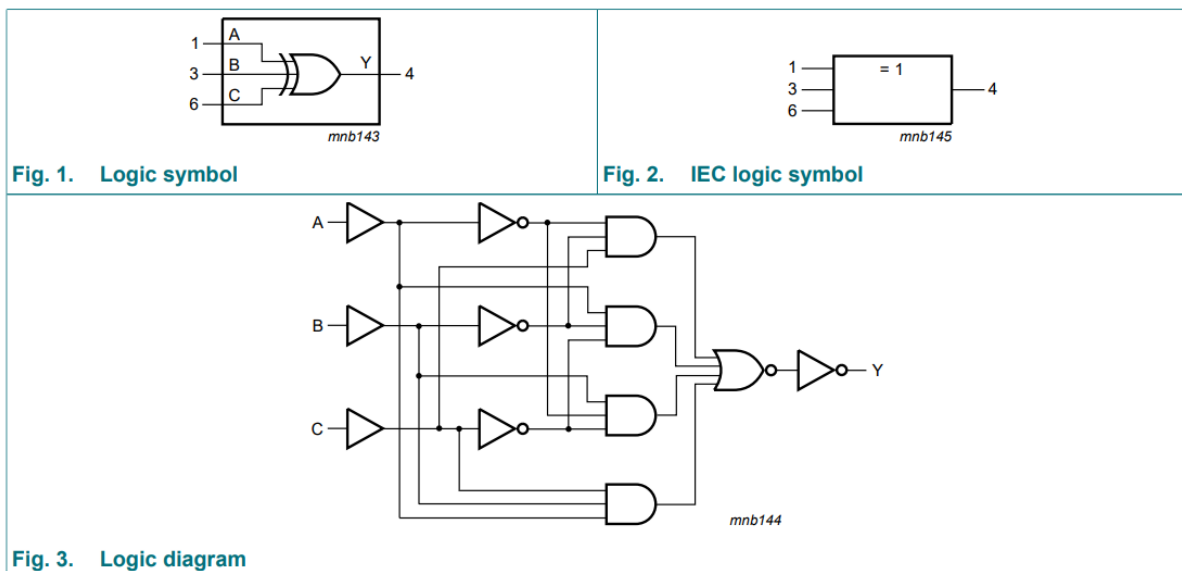
An example prescaler schematic [4]



WEEK 4

● How does a multiple input XOR gate work?

The first two signals are fed into an XOR gate, then the output of that gate is fed into a second XOR gate together with the third signal, and so on for any remaining signals. The result is a circuit that outputs a 1 when the number of 1s at its inputs is odd, and a 0 when the number of incoming 1s is even. This makes it practically useful as a parity generator or a modulo-2 adder. 3Exclusive OR is a logic element having three inputs and one output, the level “log. 1” on which appears, provided that the level “log. 1” is present only at one of its inputs. The 74LVC1G386 microchip implements a parity generator using a three-input XOR gate. The following tables are from the datasheet[5].



H = HIGH voltage level; L = LOW voltage level.

Input			Output
A	B	C	Y
L	L	L	L
L	L	H	H
L	H	L	H
L	H	H	L
H	L	L	H
H	L	H	L
H	H	L	L
H	H	H	H

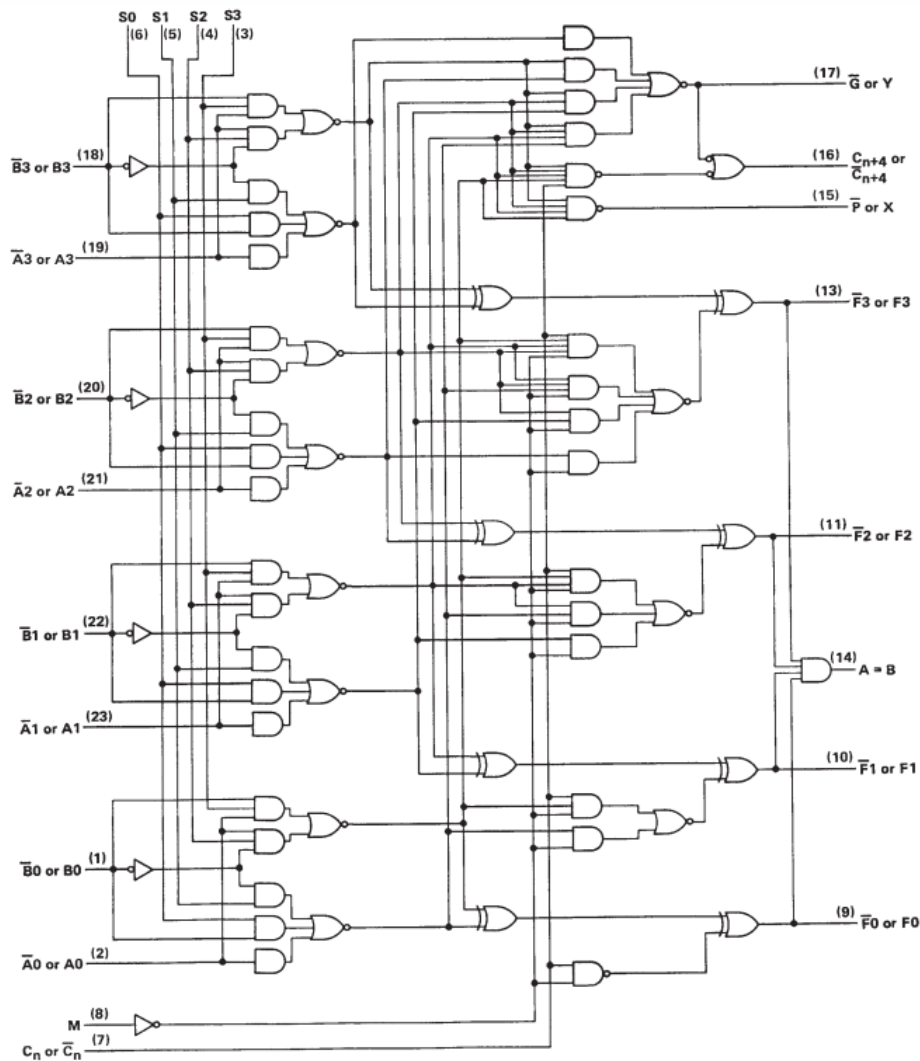
● Research ALU designs for different CPUs.

74181 is cascadable ALU of the 1960s and first of the kind. ALU operation and complexity is better understood by the features of 74181, although much more is offered by modern ALUs. Even today 74181 is of academic interest in teaching Computer architecture.

- 4-bit Arithmetic and Logical Unit for fixed-point operations
- Inputs: Two operands A and B of 4-bit width
- Output: F 0-3 (4 bit width)
- Mode selection with M - defines Arithmetic or logical mode
- Function select with 4 lines (S0-3)
- 16 sets of Arithmetic and 16 Logical Operations possible (as detailed in Function table)
- Carry in used as a special input; Cin disabled for logical operations
- Look Ahead Carry Adder Principle employed for faster output propagation with P, G outputs.

Following is the function table and logic diagram of the 74181 ALU design

Mode Select Inputs				Active LOW Operands & F _n Outputs		Active HIGH Operands & F _n Outputs	
S3	S2	S1	S0	Logic (M = H)	Arithmetic (Note 2) (M = L) (C _n = L)	Logic (M = H)	Arithmetic (Note 2) (M = L) (C _n = H)
L	L	L	L	$\bar{A}$	A minus 1	$\bar{A}$	A
L	L	L	H	$\bar{A}\bar{B}$	AB minus 1	$\bar{A} + \bar{B}$	A + B
L	L	H	L	$\bar{A} + \bar{B}$	$\bar{A}\bar{B}$ minus 1	$\bar{A} B$	A + $\bar{B}$
L	L	H	H	Logic 1	minus 1	Logic 0	minus 1
L	H	L	L	$\bar{A} + \bar{B}$	A plus (A + $\bar{B}$)	$\bar{A}\bar{B}$	A plus $\bar{A}\bar{B}$
L	H	L	H	$\bar{B}$	AB plus (A + $\bar{B}$)	$\bar{B}$	(A + B) plus $\bar{A}\bar{B}$
L	H	H	L	$\bar{A} \oplus \bar{B}$	A minus B minus 1	A $\oplus$ B	A minus B minus 1
L	H	H	H	A + $\bar{B}$	A + $\bar{B}$	$\bar{A}\bar{B}$	AB minus 1
H	L	L	L	$\bar{A} B$	A plus (A + B)	$\bar{A} + B$	A plus AB
H	L	L	H	A $\oplus$ B	A plus B	$\bar{A} \oplus \bar{B}$	A plus B
H	L	H	L	B	$\bar{A}\bar{B}$ plus (A + B)	B	(A + $\bar{B}$) plus AB
H	L	H	H	A + B	A + B	AB	AB minus 1
H	H	L	L	Logic 0	A plus A (Note 1)	Logic 1	A plus A (Note 1)
H	H	L	H	$\bar{A}\bar{B}$	AB plus A	A + $\bar{B}$	(A + B) plus A
H	H	H	L	AB	$\bar{A}\bar{B}$ minus A	A + B	(A + $\bar{B}$) plus A
H	H	H	H	A	A	A	A minus 1



Some common features and characteristics of ARM ALUs include:

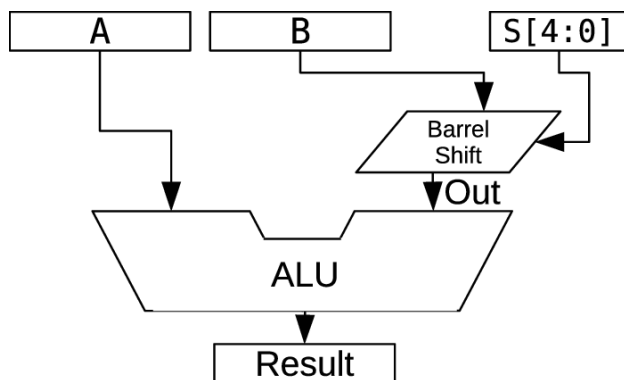
Barrel Shifter: ARM ALUs often include a barrel shifter, which allows for efficient shifting and rotating of data. This feature is particularly useful for bit manipulation operations and is commonly found in ARM processors.

Support for Thumb Instruction Set: Many ARM processors support the Thumb instruction set, which includes instructions that enable code density and efficiency. The ALU design may include optimizations for executing Thumb instructions efficiently.

Vector Processing and SIMD Operations: Some ARM processors feature ALUs optimized for vector processing and SIMD (Single Instruction, Multiple Data) operations. These ALUs are designed to handle parallel computations on multiple data elements simultaneously, which is useful for tasks like multimedia processing and signal processing.

Floating-Point Unit (FPU): ARM processors used in applications requiring floating-point arithmetic often include a separate Floating-Point Unit (FPU). The FPU works in conjunction with the ALU to perform floating-point operations efficiently.

Scalar ALU Operations: Alongside vector and SIMD operations, ARM ALUs support traditional scalar arithmetic and logical operations. These operations are essential for general-purpose computing tasks.



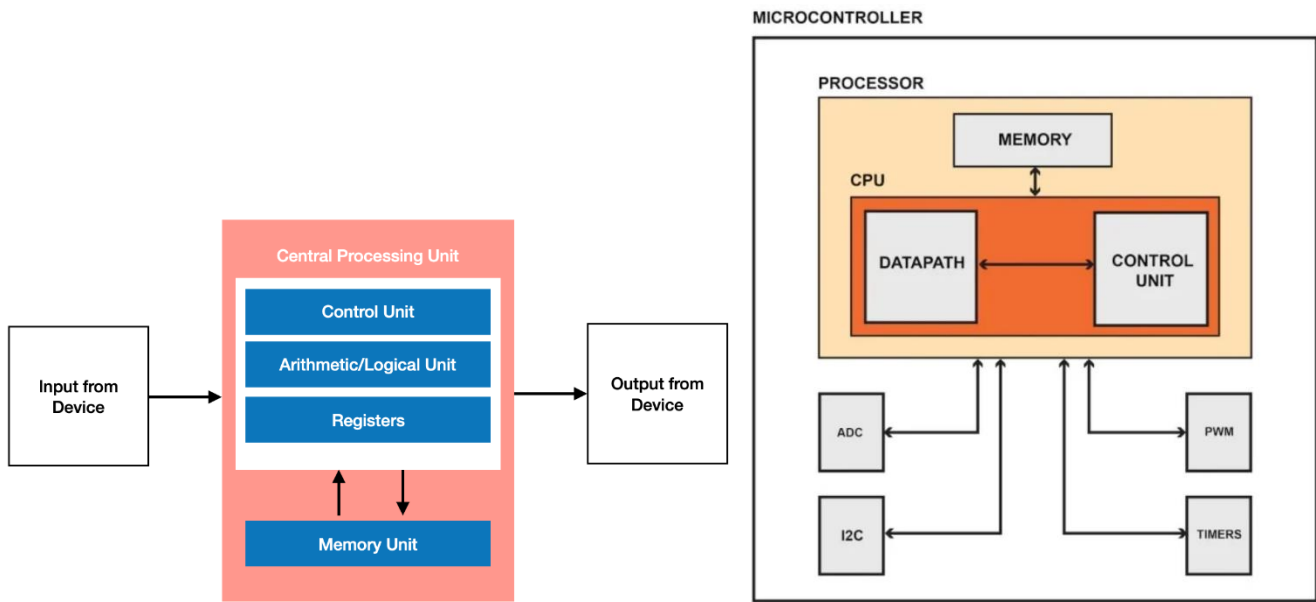
Schematic on the left is a simple representation of a basic Arithmetic Logic Unit in ARM-Cortex Series

● What's the difference between a CPU and an MCU?

CPUs typically have a more complex architecture with multiple execution units, caches, pipelining, and support for out-of-order execution. They often have higher clock speeds and are designed to handle a wide variety of computational tasks efficiently. The general-purpose computer that we use are using this type of architecture. CPUs run operating systems, handles multiple tasks simultaneously (multitasking), and supports a wide range of input/output devices.

MCUs are designed with a simpler architecture optimized often used for specific control and processing tasks within a system for embedded applications. They usually have a single-core processor, limited memory and storage, and basic input/output (I/O) peripherals integrated on the same chip. They are programmed to execute dedicated firmware or software to control the behavior of the system, interact with sensors and actuators, and perform tasks such as data logging, signal processing, motor control, and more. The architecture of an MCU is tailored for real-time control and low-power operation.

Below, there is a simple diagram on CPU (on the right) and MCU (left)



- Write the assembly programs doing the same operation with the following C codes, using 8086 instruction set on EMU8086.

```

sum DW 0
i DW 0
MOV AX, DATA
MOV DS, AX

MOV CX, 11
MOV AX, 0
MOV i, AX    ; i is set to 0

LOOP_START:
    ADD AX, i    ; Add i to sum
    INC i        ; Increment i
    LOOP LOOP_START ; Repeat until CX becomes 0
    MOV sum, AX
  
```

MOV AX, DATA

MOV DS, AX

MOV CX, 5 ; Set CX to 5 for outer loop (i)

MOV SI, 0 ; Set SI to 0 for outer loop counter

FIRST_LOOP:

INC SI

MOV BX, 5

MOV DI, 0

SECOND_LOOP:

INC DI

MOV DX, 5 ; Set DX to 5 for third loop

MOV BP, 0

THIRD_LOOP:

INC BP

INC WORD PTR [t]

DEC DX ; Decrement DX (k counter)

JNZ THIRD_LOOP ; If DX is not zero, repeat innermost loop

DEC BX ; Decrement BX (j counter)

JNZ SECOND_LOOP ; If BX is not zero, repeat second loop

DEC CX ; Decrement CX (i counter)

JNZ FIRST_LOOP ; If CX is not zero, repeat first loop

.DATA

t DW 0

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